AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 28 · STUDENT LESSON

Privacy by Design

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Name the purpose. Minimise the fields. Verify every declared copy.

Design bounded data retention and distinguish requested cleanup from verified store outcomes.

Keep one habit: Data checklist → three-day folder/cache trace → field audit → seven-day practice → independent five-day log case.

MAIN LESSON · SECTION 1

Which data does the task need, and where can copies remain?

> Privacy choices belong in fields, access rules, lifetimes and deletion evidence.

Your learning goals

I can explain the **mechanism** and terminology of data **retention** and deletion.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall state scope and disclosure paths

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Conversation State and Memory and Red Team the Product: keep only task-relevant state, define its lifetime and inspect actual data paths. A hidden copy can survive a visible reset.

Your end goal

By the end, I can explain privacy by design using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Requests Responses and JSON. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 28: Privacy by Design

Data minimization, **retention** rules, and **consent** belong in the architecture.



CREATOR MISSION

Remove unnecessary student data from the design.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Minimise first, then trace the data you retain

> A deleted screen row is not evidence that every copy was removed.

Connect purpose, fields and stores

Privacy by design starts before collection: name the task purpose and decide whether each field is necessary. Less unnecessary content means fewer copies to protect and delete. For kept data, define who may access it, where it is copied, its retention boundary and what confirms deletion. A permission checkbox does not make unrelated collection useful.

These classroom policies are fictional design exercises, not legal retention rules or tests of a real service.

Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Use fictional records only

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use privacy by design to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Write the field plan and exact retention boundary

> Do not mix thresholds from separate exercise policies.

The complete seven/three/five-day policy map

Keep the policy labels separate. W2 uses a seven-day content policy: delete when age >= 7 days in primary store and cache; allowed content-free audit metadata may remain until age >= 30 days. A backup expiry process is not supplied, so report its status unknown and request the policy rather than promise complete deletion. The main pictured case uses a different three-day policy for its declared folder/cache stores.

The independent case states five days. These are not universal settings.

A purpose-to-deletion checklist

Data-design question	Evidence to write
Purpose and necessity	Which user task requires the field? Can it be omitted or shortened?
Permission and access	What approved use and access role apply? Missing permission is not inferred.
Copies and derived data	Primary store, cache, log, export and declared backup locations

Data-design question	Evidence to write
Retention boundary	Age definition, exact threshold and inclusive/exclusive rule
Deletion verification	Record/store IDs, result, residual copies and unresolved policies
Minimal audit	Allowed event metadata without retained content; metadata still needs purpose and access rules

Original W2 policy and boundary

Toy policy retains session content for 7 days and deletes it when age is at least7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Explain the deciding check

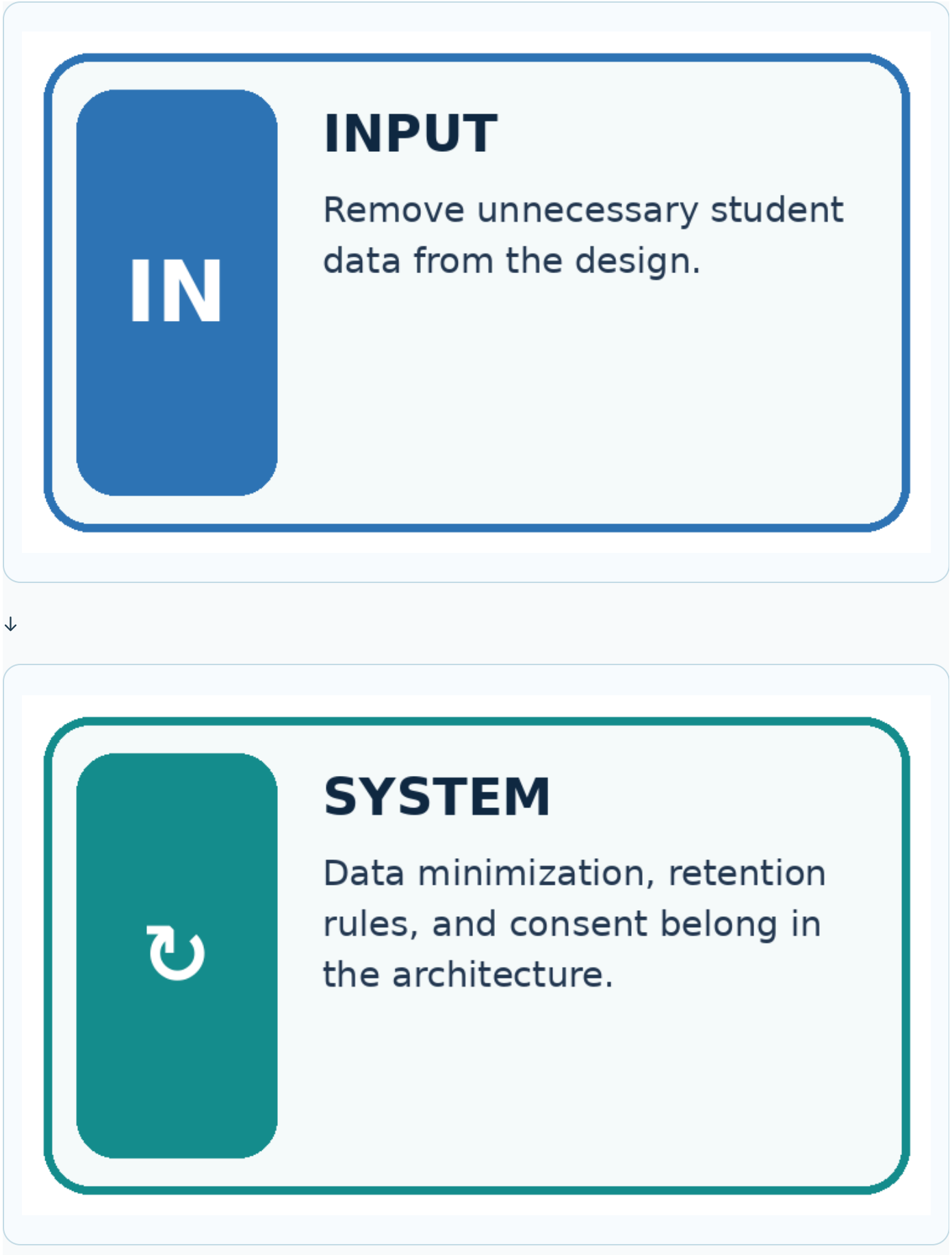
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Remove unnecessary student data from the design.
↓
SYSTEM
Data minimization, retention rules, and consent belong in the architecture.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Data retention and deletion — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Record ages are6, 7, 9 days. Cache contains copies of all three.	Toy policy retains session content for 7 days and deletes it when age is at least7.	Retain age6; delete content ages7 and9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry process.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

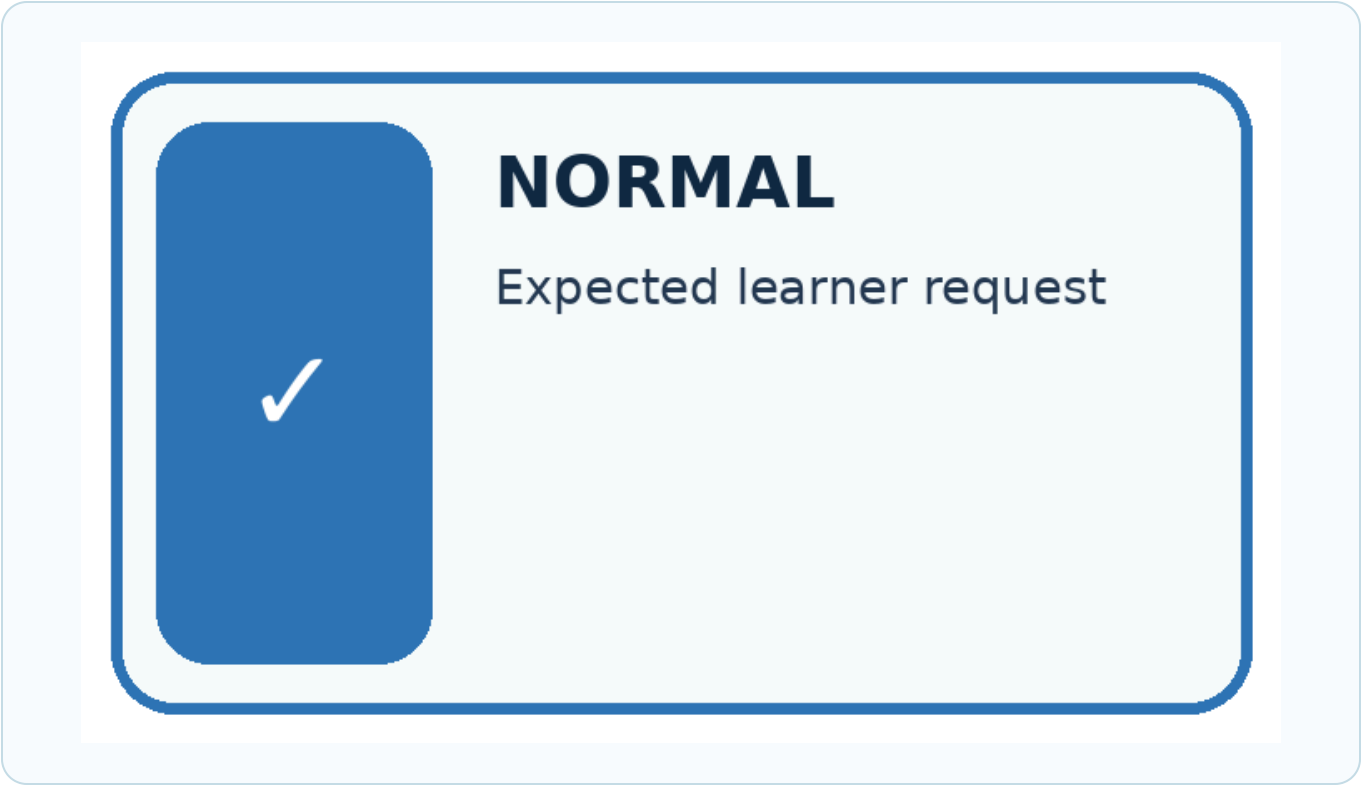
Connect the picture and the explanation

MAIN LESSON · SECTION 4

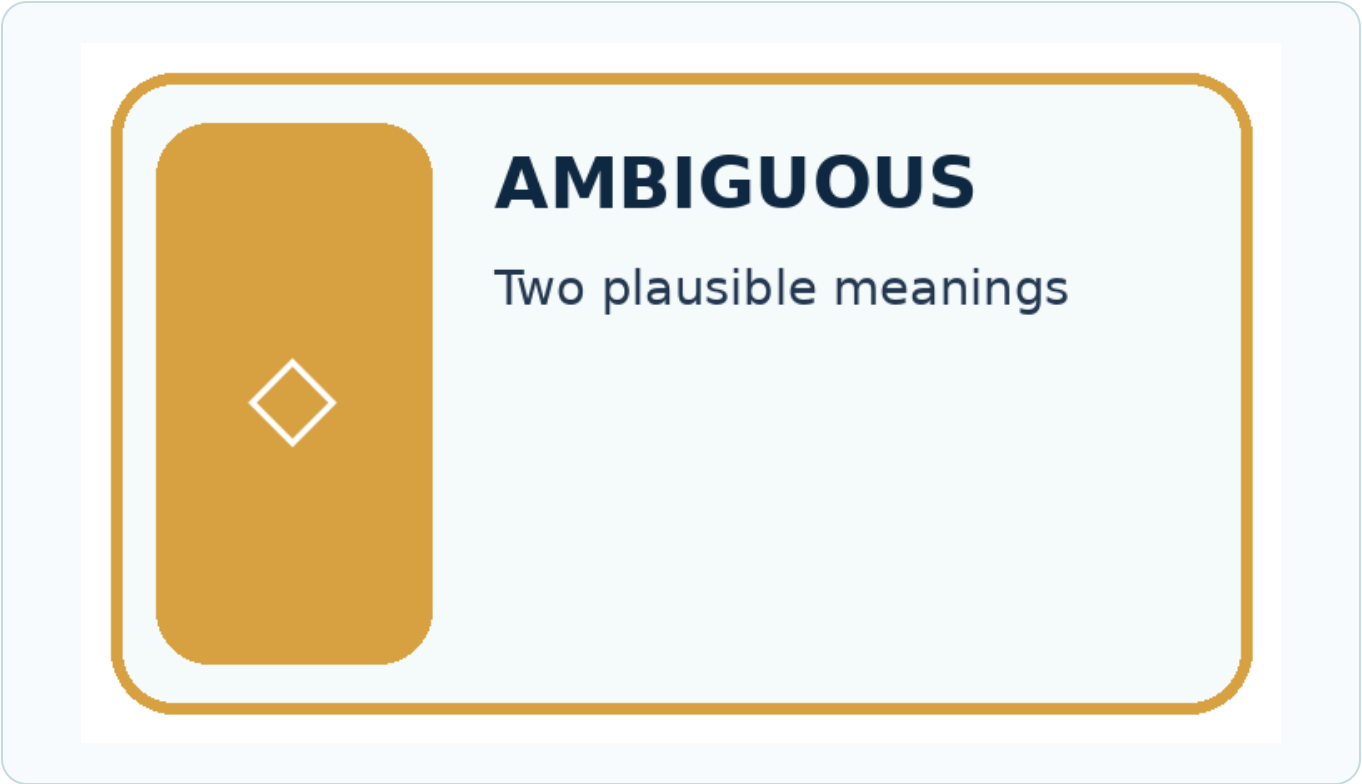
Compare intended deletion with evidence across copies

> A known remaining copy makes a deletion claim incomplete.

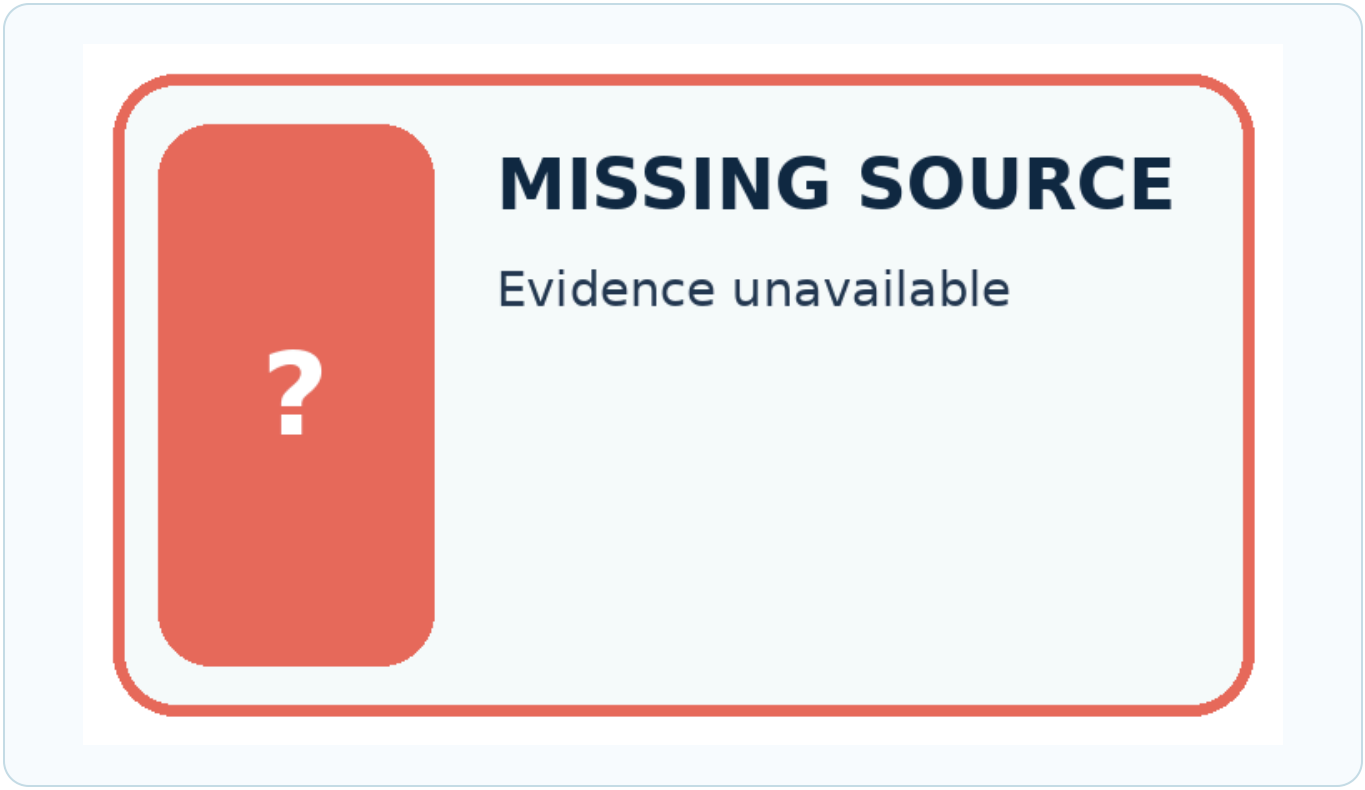
Four pictures with a purpose



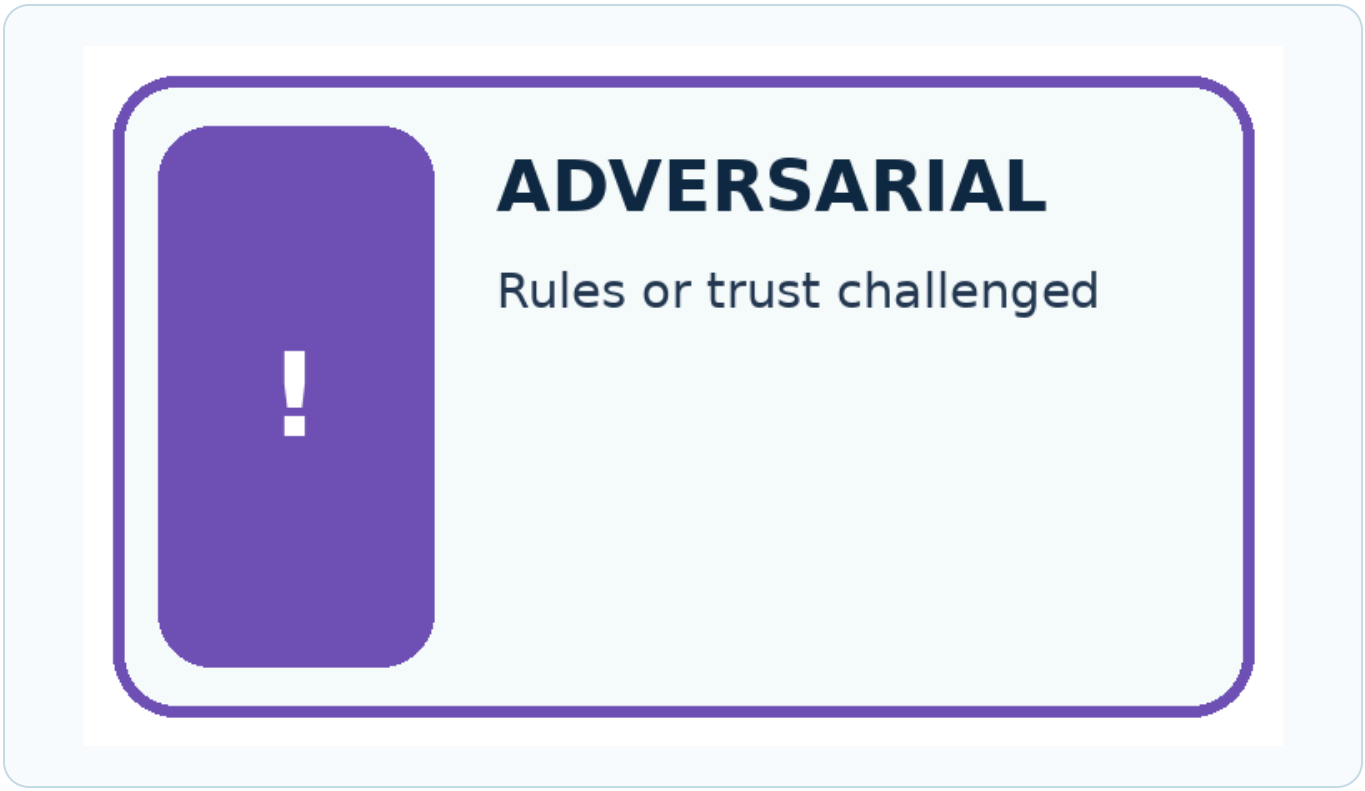
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: The cache deletion fails.

▼ Check after predicting

Prediction: Report incomplete deletion rather than success.

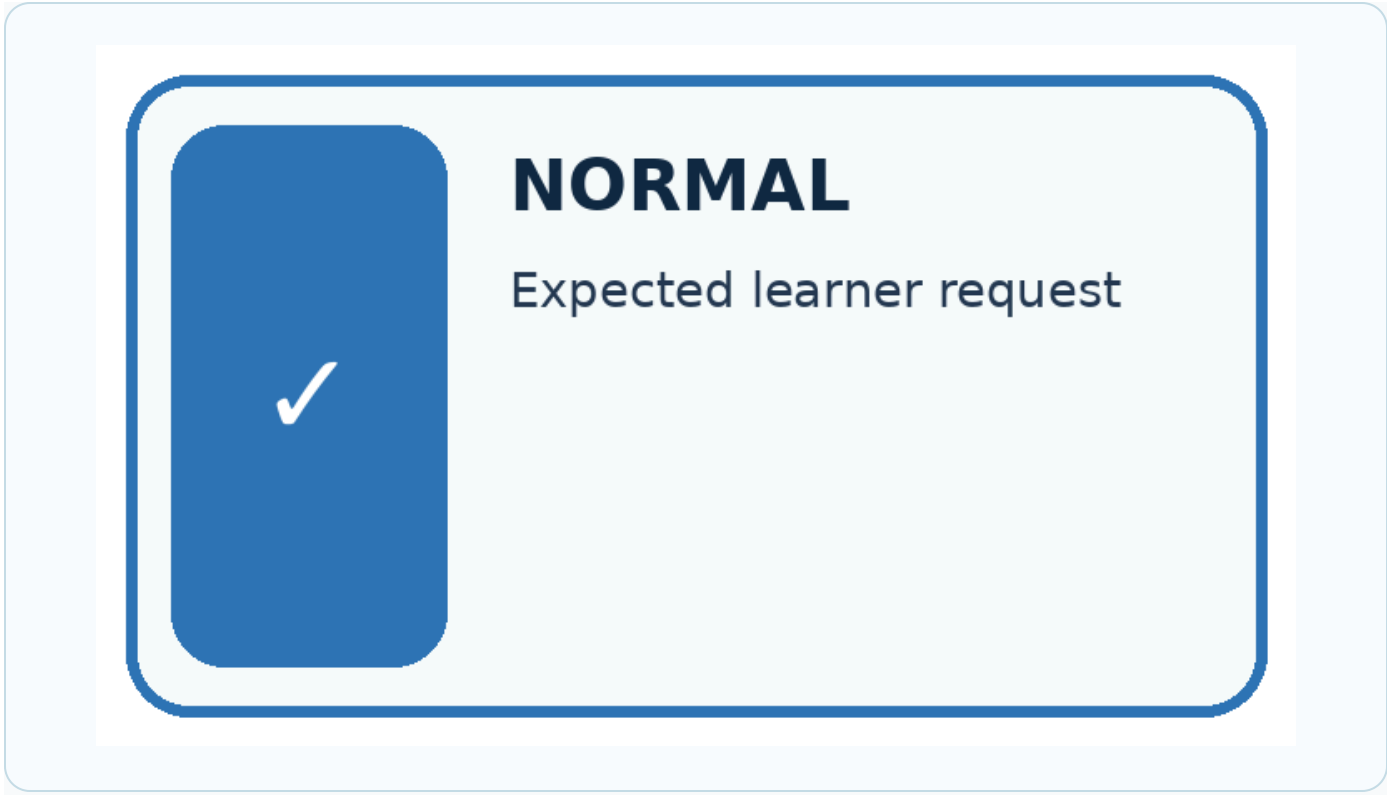
Reason: Removing the working copy does not remove the known retained cache copy.

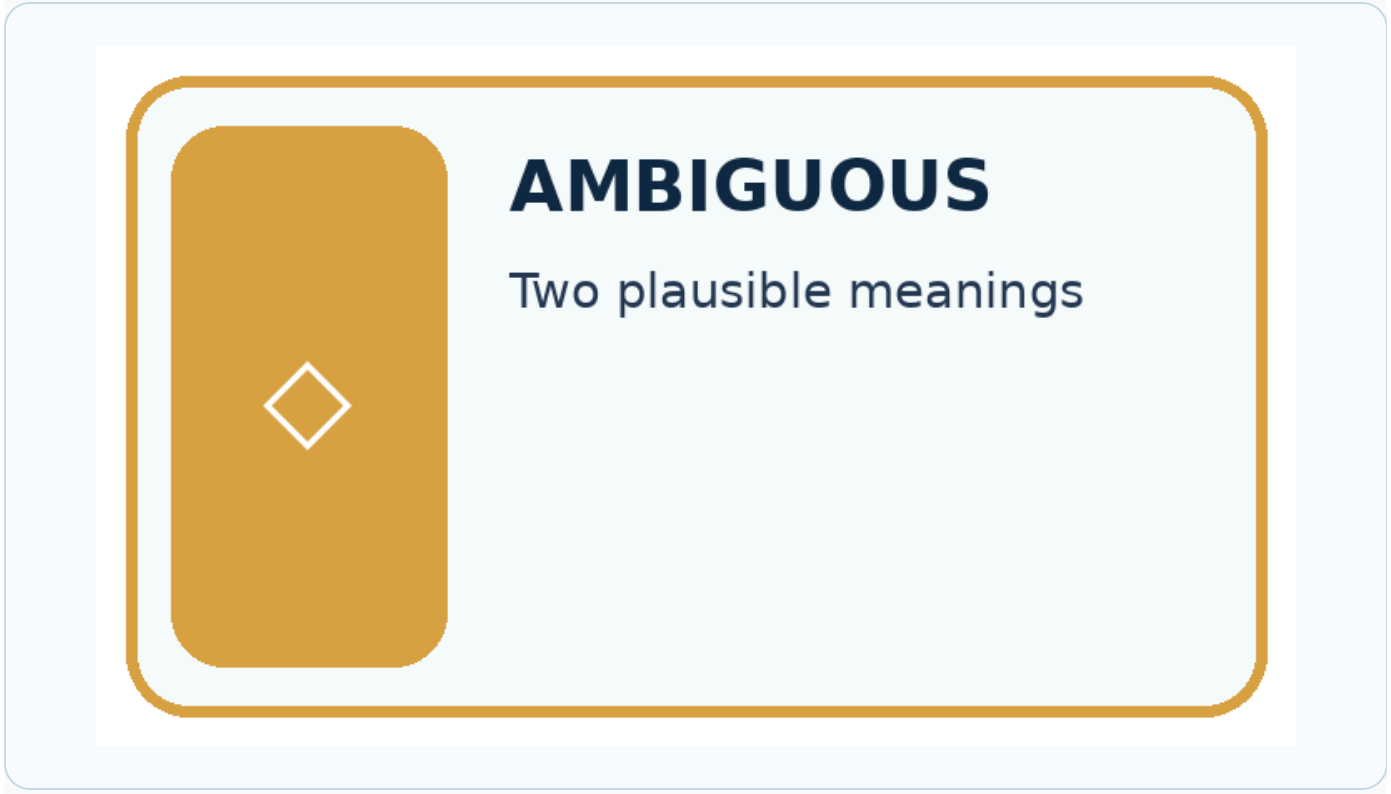
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the three-day folder-and-cache case

> At the inclusive boundary, B must be removed from both declared locations.

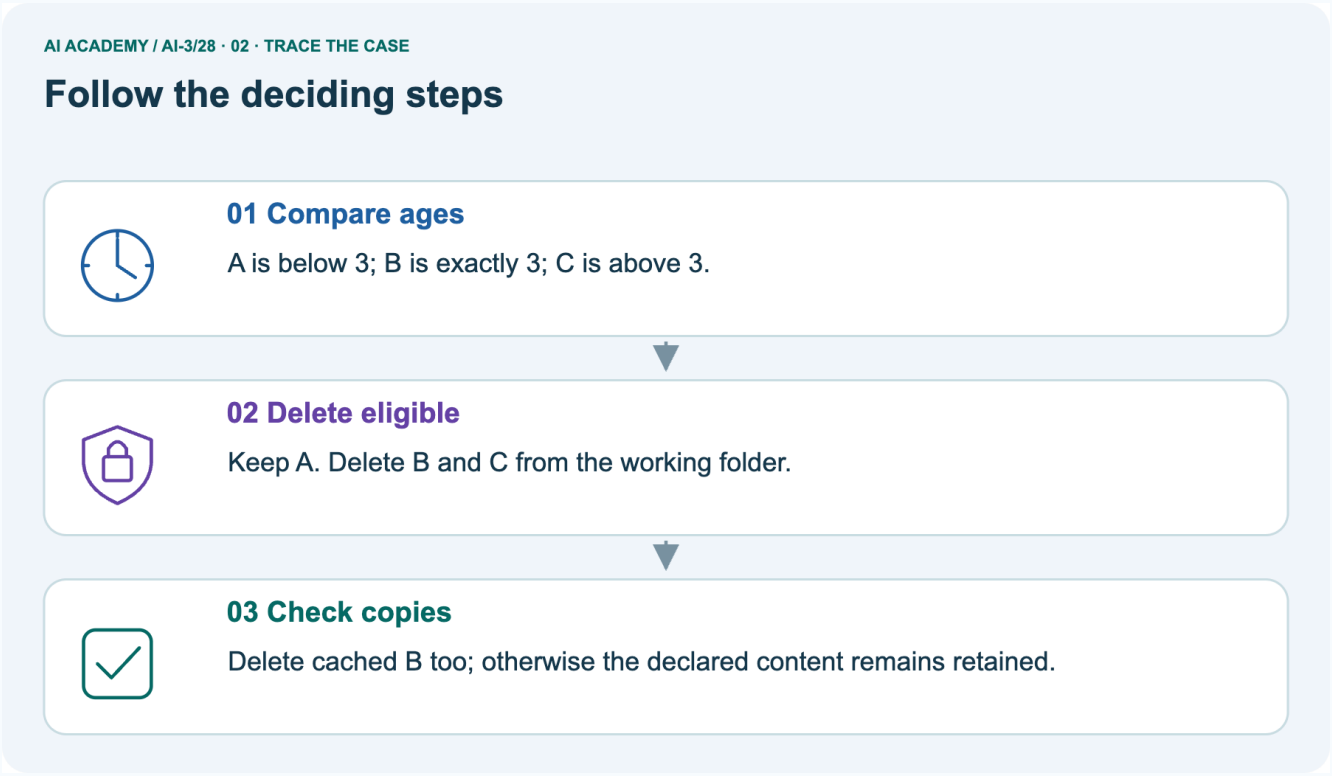
Read the complete A/B/C policy and stores

Trace deletion through both a working folder and a cache.

Fictional **retention** policy: delete records aged at least 3 days. Ages A=2, B=3, C=4; the cache contains a copy of B.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Trace keep/delete decisions and cached B



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare ages	A is below 3; B is exactly 3; C is above 3.
2. Delete eligible	Keep A. Delete B and C from the working folder.
3. Check copies	Delete cached B too; otherwise the declared content remains retained.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

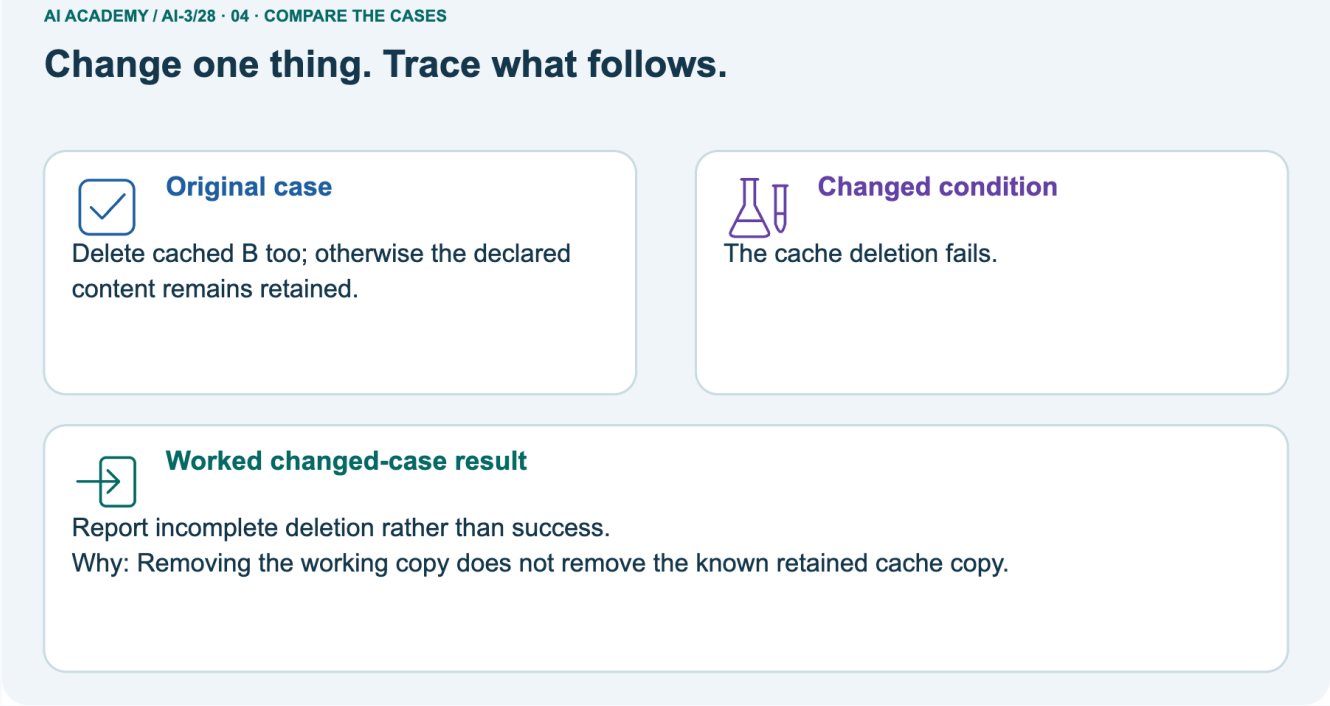
Read the picture in order. Delete cached B too; otherwise the declared content remains retained. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict failed cache deletion

- Change one thing: The cache deletion fails. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State the fictional-policy limit

- This fictional policy is a tracing exercise, not a statement of legal **retention** requirements.
- Read the labelled picture: Which IDs and locations must appear in the deletion **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate practice · Seven-day W2

Record ages are6, 7, 9 days. Cache contains copies of all three.

Retain age6; delete content ages7 and9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

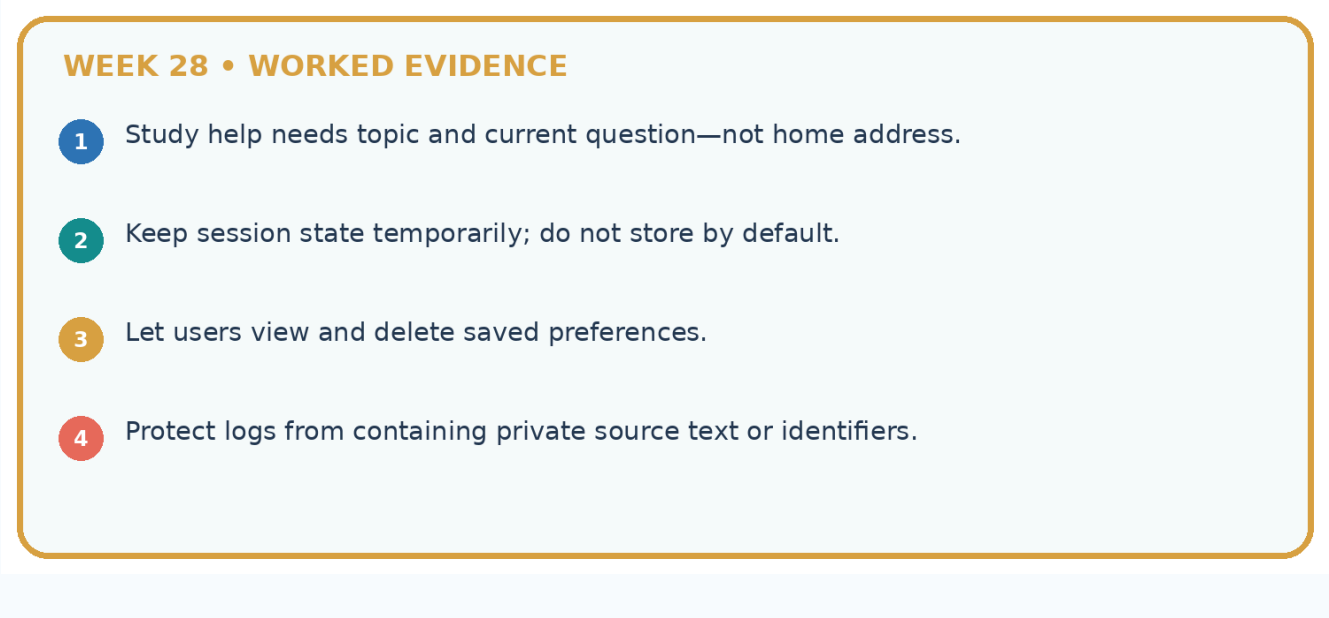
Use only fictional classroom records. Preserve the policy, record IDs, store locations and deletion outcomes needed to explain the trace; do not retain expired personal content merely to prove that deletion was requested. Report known residual copies and unspecified backup policies separately.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Study help needs topic and current question—not home address.

2. Keep session state temporarily; do not store by default.

3. Let users view and delete saved preferences.

4. Protect logs from containing private source text or identifiers.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

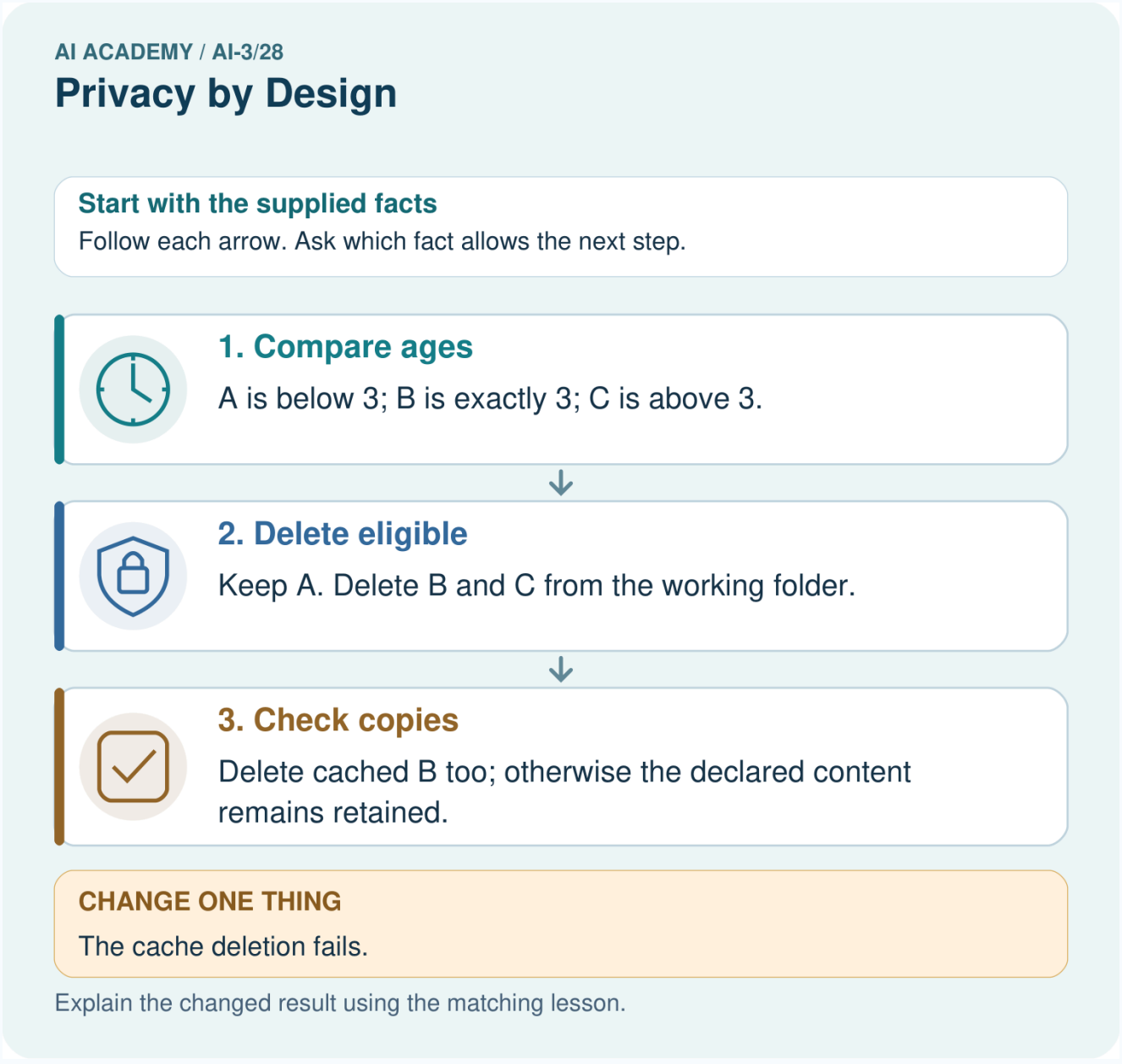
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Privacy by Design

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a visible action mistaken for a complete data lifecycle

> The evidence should expose residual copies rather than hide them.

Repair the folder-only shortcut

Claim to repair: B disappeared from the folder, so all deletion is complete. The known cached B remains if cache deletion fails. Record incomplete deletion and the remaining store; retry or escalate under the declared policy without inventing a successful result. Do not make a full-content debug log to prove deletion, because that creates another retained copy.

Likewise, unspecified backup expiry remains unknown rather than automatically successful.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Privacy can be added after the product works. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the seven-day W2 policy with less help

> Apply its own inclusive threshold and declared audit limit.

The policy and legal-scope limit

This fictional **retention** exercise is not a statement of legal **retention** requirements.

Complete the ages 6/7/9 W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Data **retention** and deletion”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit an all-future-cases claim

- Use the worked case for Data **retention** and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the pictured store evidence

- Which IDs and locations must appear in the deletion **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Explain the picture

My explanation using a pictured step and the given facts

Keep the independent five-day case separate

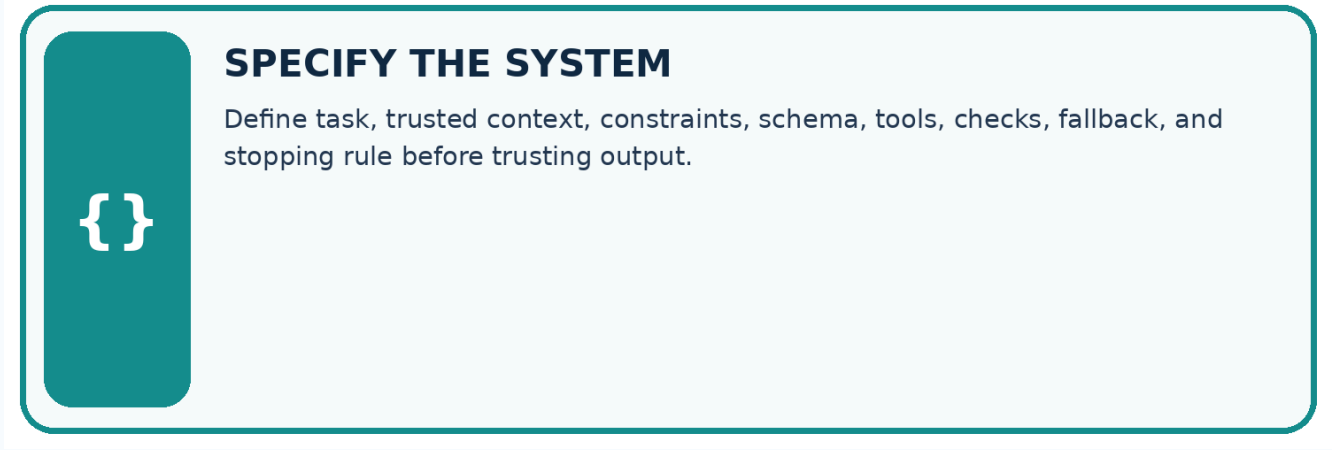
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Audit six complete field and copy decisions

A useful data plan includes both omission and lifecycle checks.

All supplied proposed fields and stores

Complete fictional proposed field/store	Purpose and design decision to audit
Selected book ID	Needed to choose this session’s reading; keep only for the session, clear on exit.
Preferred explanation length	Optional session setting; ask whether it helps the task and allow correction/reset.
Home address	No role in this glossary/reading task; omit rather than collect. No address value is supplied.
Raw chat in debug log	Full-content logging is not needed for the stated toy task; use permitted minimal event metadata instead.
Deleted primary content still in cache	Declared seven-day cleanup must cover expired cache copies too.

Complete fictional proposed field/store	Purpose and design decision to audit
Backup copy mentioned, no expiry policy	Record unresolved expiry and request the policy; no claim of complete removal.

Audit, minimise and trace fictional cleanup

1. Audit every supplied design row for purpose, necessity, permitted use/access, lifetime, copies and deletion evidence. The first two rows describe proposed session settings; the cleanup rows use the separate seven-day example. Remove the unnecessary address field from the design, not a real file. Replace full-content logging with a minimal declared audit plan and state its own access/expiry rules.

   Trace ages 6, 7, 9 in primary and cache: keep 6, remove 7 and 9; allowed content-free audit expires at 30. Backup expiry is unspecified. Simulate a cache failure and report residual content rather than success. No real pupil data, legal policy or storage deletion is involved. Keep the independent five-day case withheld.

Keep minimal non-content evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → RUN → VERIFY → EXPLAIN

Audit every field for purpose, permission, retention, access, and deletion; remove or shorten anything not essential.

Audit the six supplied fictional field/store rows for purpose, necessity, permission, access, **retention** and deletion evidence. Omit unrelated address collection, avoid full-content logs, trace primary/cache thresholds and preserve unspecified backup expiry as unknown. No real record is collected or deleted.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: trace the fresh five-day policy and content log

> Use its stated threshold and report unresolved store outcomes.

Independent records and full store scope

Retention is5 days, inclusive **deletion boundary**. Ages4, 5, 6; a log stores complete content. Trace required action.

Apply the stated five-day policy inclusively: deletion is due when age >= 5 days. Logs are a declared store too. For this toy contract, full-content logs are not permitted audit metadata; identify required handling and residual copies rather than silently keeping them. No backup rule or actual deletion result is supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A practice quiz needs a topic and difficulty. A form also asks for home address and date of birth. What should be removed and why?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Create a minimal data map and visible memory controls before expanding features.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A study tool needs topic and reading level but proposes storing addresses and all conversations forever. Design a smaller record and a deletion rule.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project’s purpose and store map visible

> Every copied value creates another lifecycle responsibility.

This week’s contribution

For the Verified Study Buddy, draw the data path before collecting values. Keep necessary session settings minimal and visible; give each store and derived copy a declared lifetime and access owner. Use content-free audit events where sufficient, and protect that metadata too. Record a deletion request separately from verified store outcomes and unresolved backups.

Do not copy private material into a demonstration or a repair report.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK’S CHECKPOINT Create a minimal data map and visible memory controls before expanding features.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

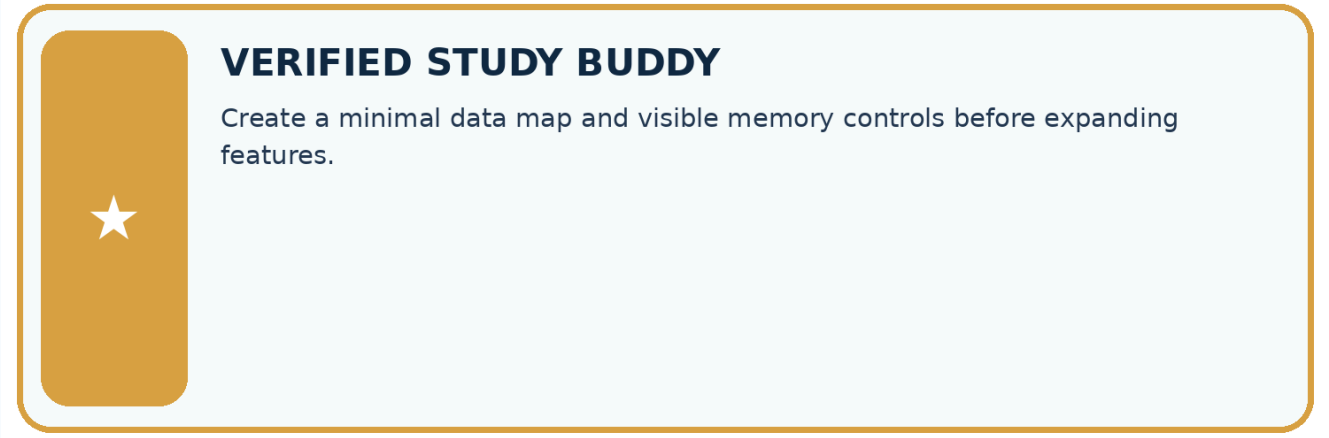
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: name the purpose, boundary and remaining copy

> Deletion evidence needs store outcomes, not just a hidden row.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of data **retention** and deletion

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
data minimization	Collecting and retaining only the information needed for a specified purpose.	
consent	An informed and freely given agreement to a specific use or action.	
retention	Keeping information for a period of time rather than deleting it immediately.	
access control	Rules and mechanisms deciding who may view information or perform an action.	

Terms for the connected explanation

Word	Meaning
Retention period	The declared lifetime of stored information

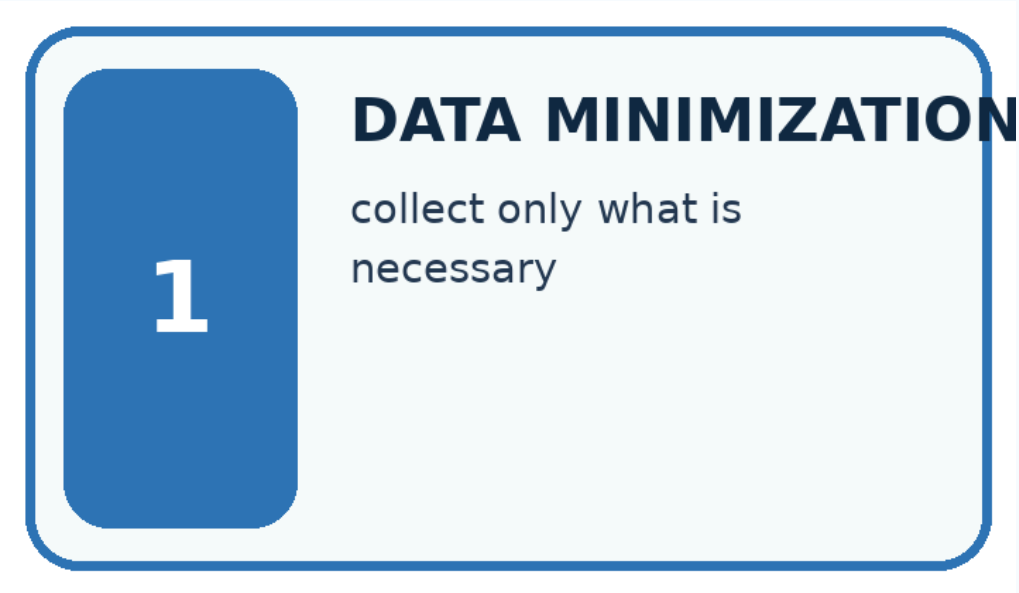
Word	Meaning
Deletion boundary	The age at which removal applies
Residual copy	A remaining duplicate in another store

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
CONSENT
informed permission for a defined use
3
RETENTION
how long data is kept



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

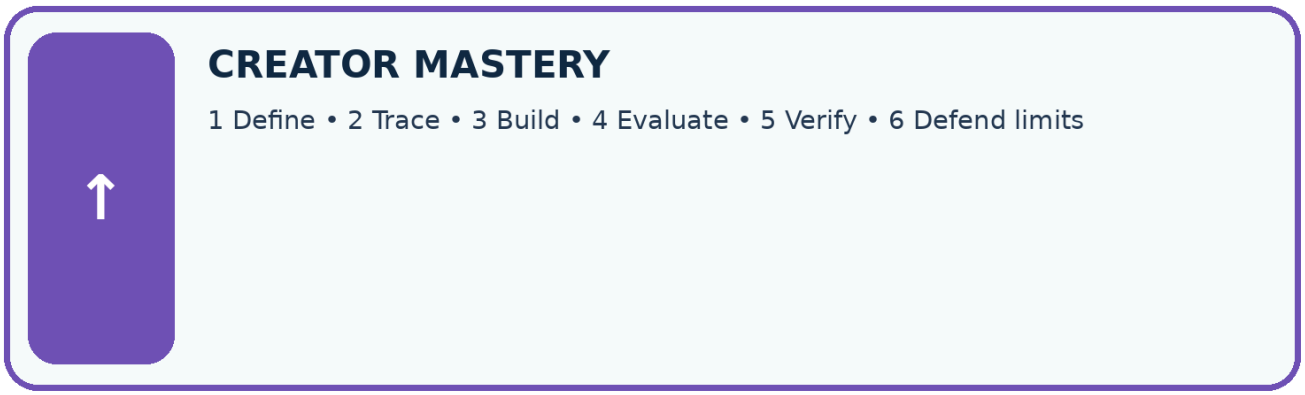
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Trace the fresh five-day lifecycle independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

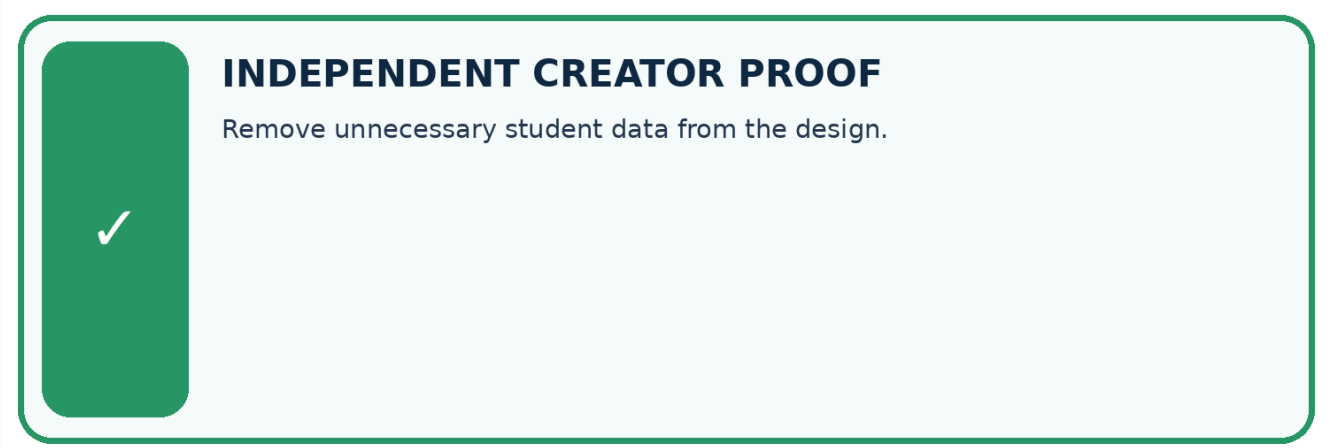
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer ‘why?’

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week’s idea

Use after practice. Trace the fresh five-day lifecycle independently is the selected independent case.



EXPLANATION CHECK Require inclusive threshold, all declared copies, minimal audit and explicit residual or unspecified stores.

Write or draw your reasoning and evidence.

▼ **Extra practice: Keep a small, purposeful data plan**

REFERENCE · SECTION 13

Extra practice: Keep a small, purposeful data plan

> Try the question before opening its worked explanation.

Read the supplied case and question

A study tool plans to retain names, addresses and all chats forever.

Design a smaller data plan.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Keep only needed preferences or progress, with **retention**, deletion and **access control**s.
Avoid unnecessary addresses and raw chat storage.

Evidence or stage	Value or result
Preference	Reading level may be useful
Address	Unnecessary for chapter study
Raw chats	Avoid indefinite retention
Controls	Define access, retention and deletion

▼ Earlier wording, alternatives and picture transcripts

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Keep a small, purposeful data plan

A study tool plans to retain names, addresses and all chats forever.

Design a smaller data plan.

Keep only needed preferences or progress, with **retention**, deletion and **access control**s. Avoid unnecessary addresses and raw chat storage.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of data retention and deletion	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Trace deletion through both a working folder and a cache.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Trace deletion through both a working folder and a cache.
- Which labelled step matches this detail? Delete cached B too; otherwise the declared content remains retained.
- 1. Compare ages
- 2. Delete eligible
- 3. Check copies
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Check copies Delete cached B too; otherwise the declared content remains retained. Read the picture in order.

  Delete cached B too; otherwise the declared content remains retained. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: The cache deletion fails. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Report incomplete deletion rather than success.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Report incomplete deletion rather than success. Removing the working copy does not remove the known retained cache copy.

My selected conclusion, deciding fact and explanation

▼ Check the changed case

Repair a missing building block

Repair and save your work

Review Red Team the Product (ai-3/27). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Quality Rubrics (ai-3/25). Goal: Judge an output using separate observable rubric criteria. Short practice: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Requests Responses and JSON (ai-3/22). Goal: Parse JSON into Python records. Short practice: Would converting the text silently prove the source count is correct?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

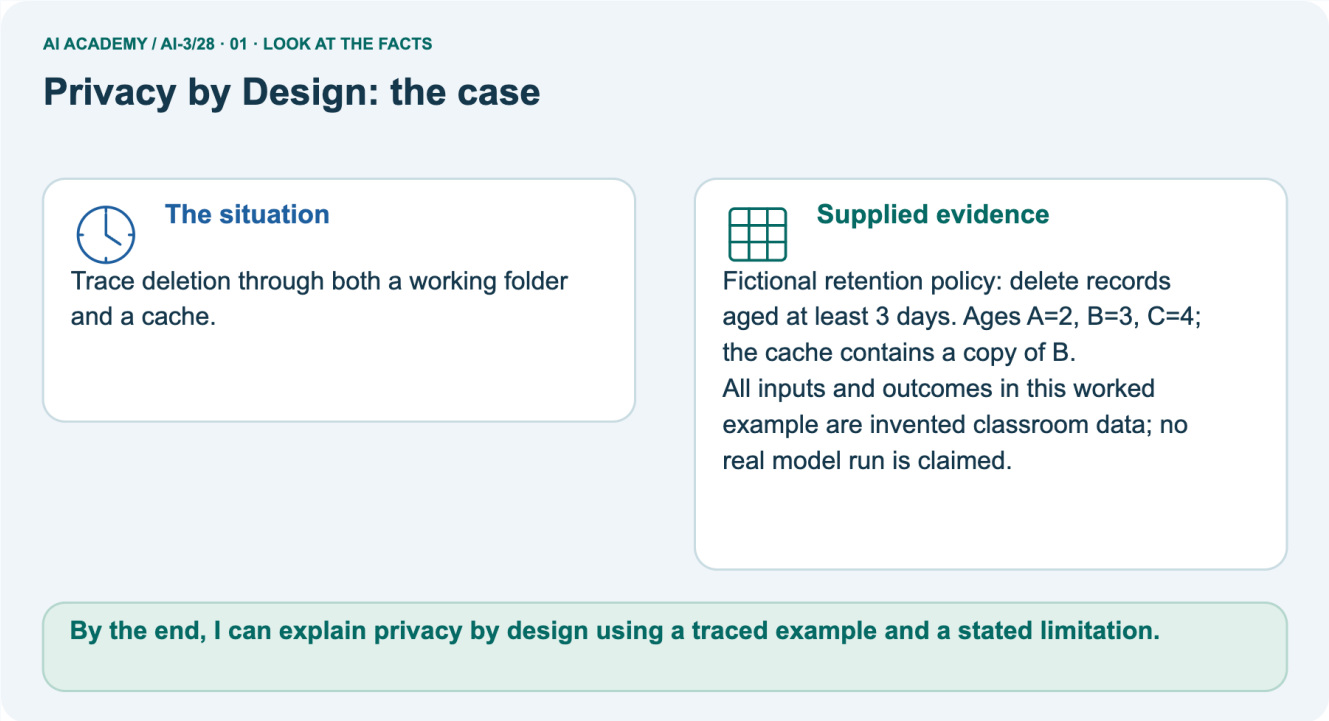
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Privacy by Design: the case



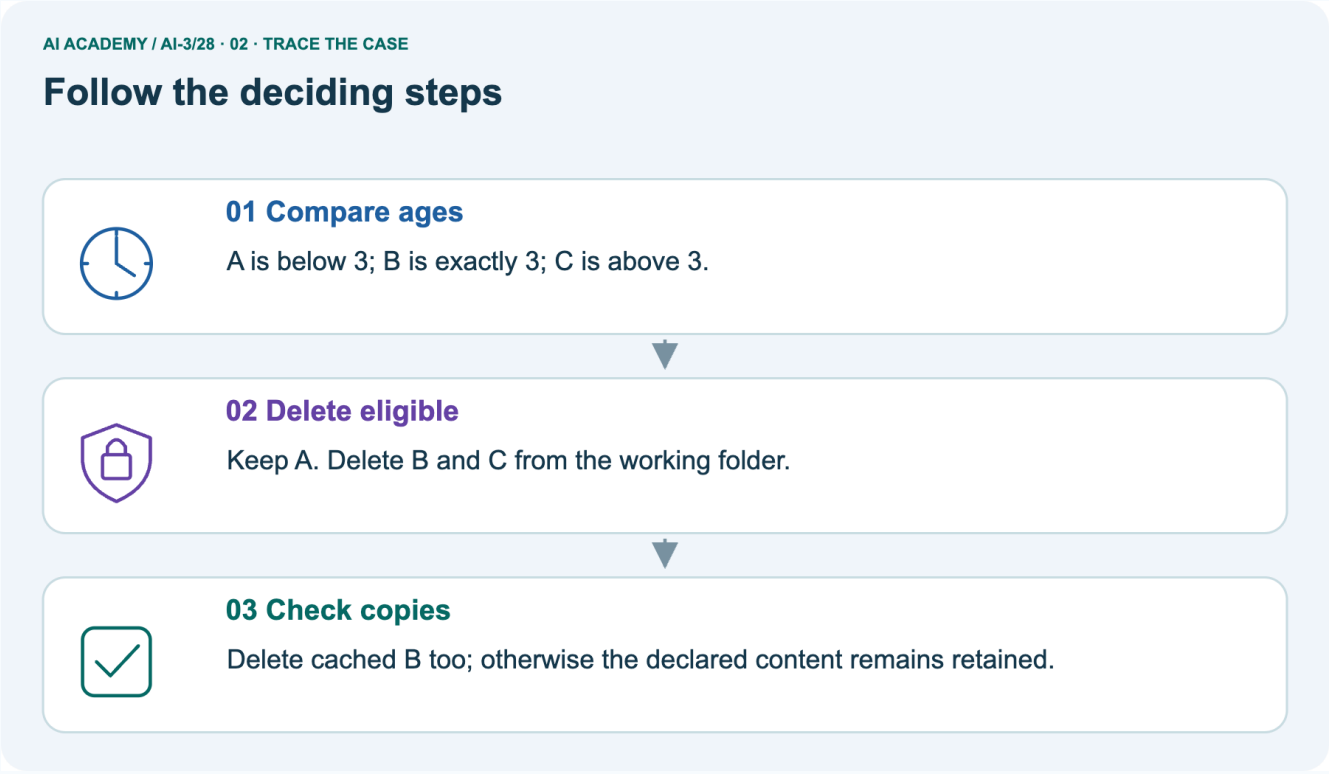
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace deletion through both a working folder and a cache. Supplied facts: Fictional **retention** policy: delete records aged at least 3 days. Ages A=2, B=3, C=4; the cache contains a copy of B.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

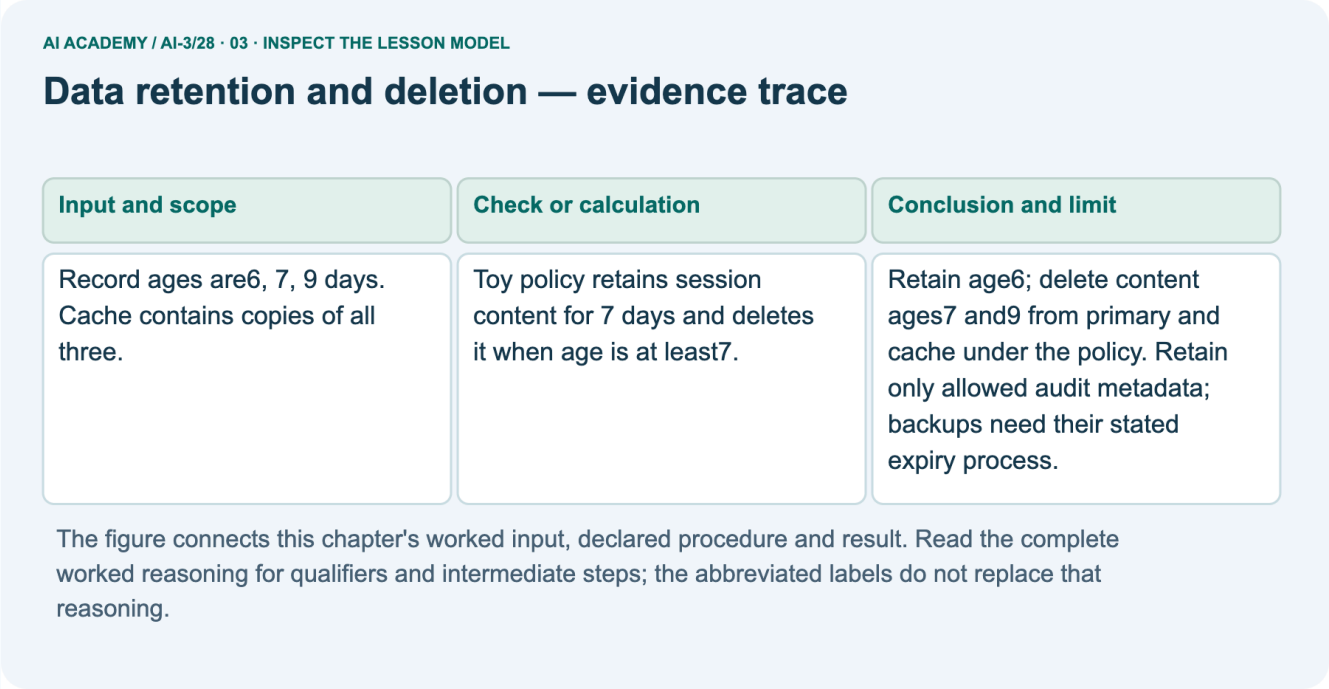


Follow the labelled picture with the matching explanation. Open the vector diagram

Compare ages: A is below 3; B is exactly 3; C is above 3. → Delete eligible: Keep A. Delete B and C from the working folder. → Check copies: Delete cached B too; otherwise the declared content remains retained.

Point to each numbered stage and explain which recorded **input** it uses.

03. Data retention and deletion — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

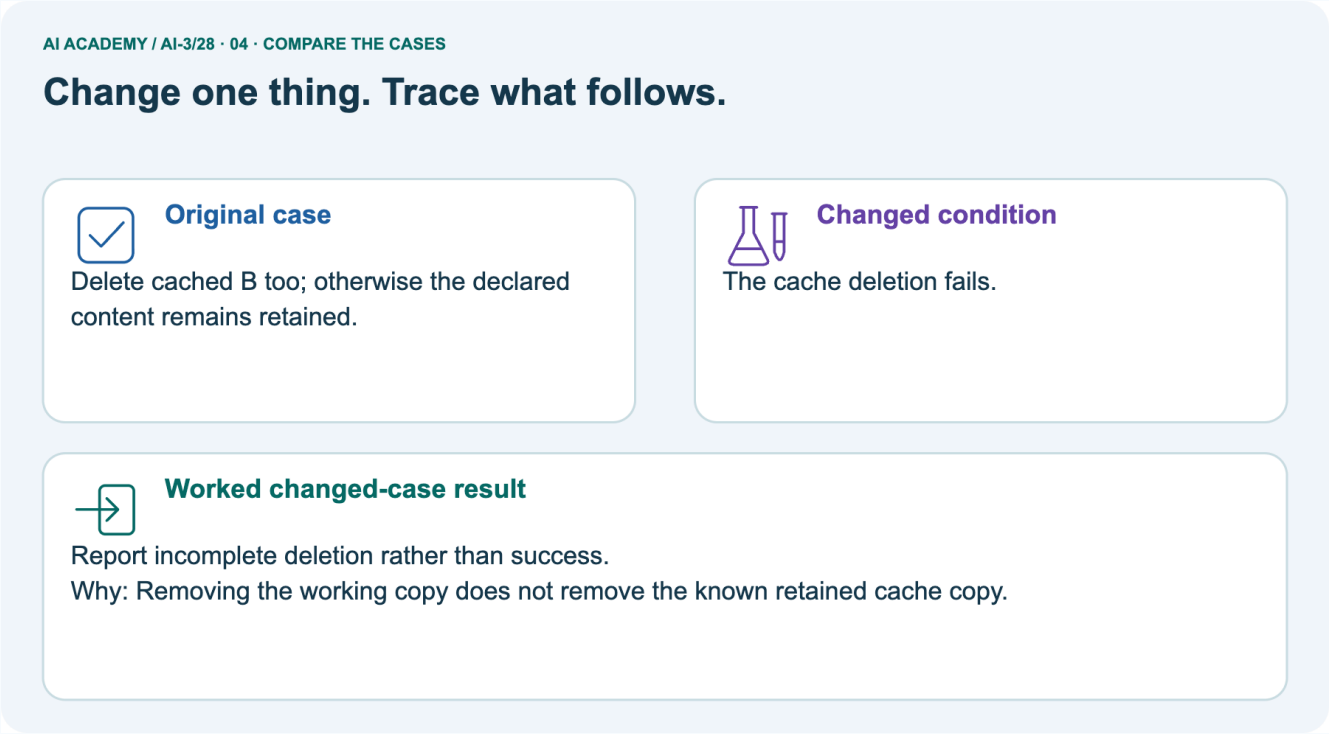
Data **retention** and deletion — **evidence** trace. Record ages are6, 7, 9 days.

Cache contains copies of all three. | Toy policy retains session content for 7 days and deletes it when age is at least7. | Retain age6; delete content ages7 and9 from primary and cache under the policy.

Retain only allowed audit metadata; backups need their stated expiry **process**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



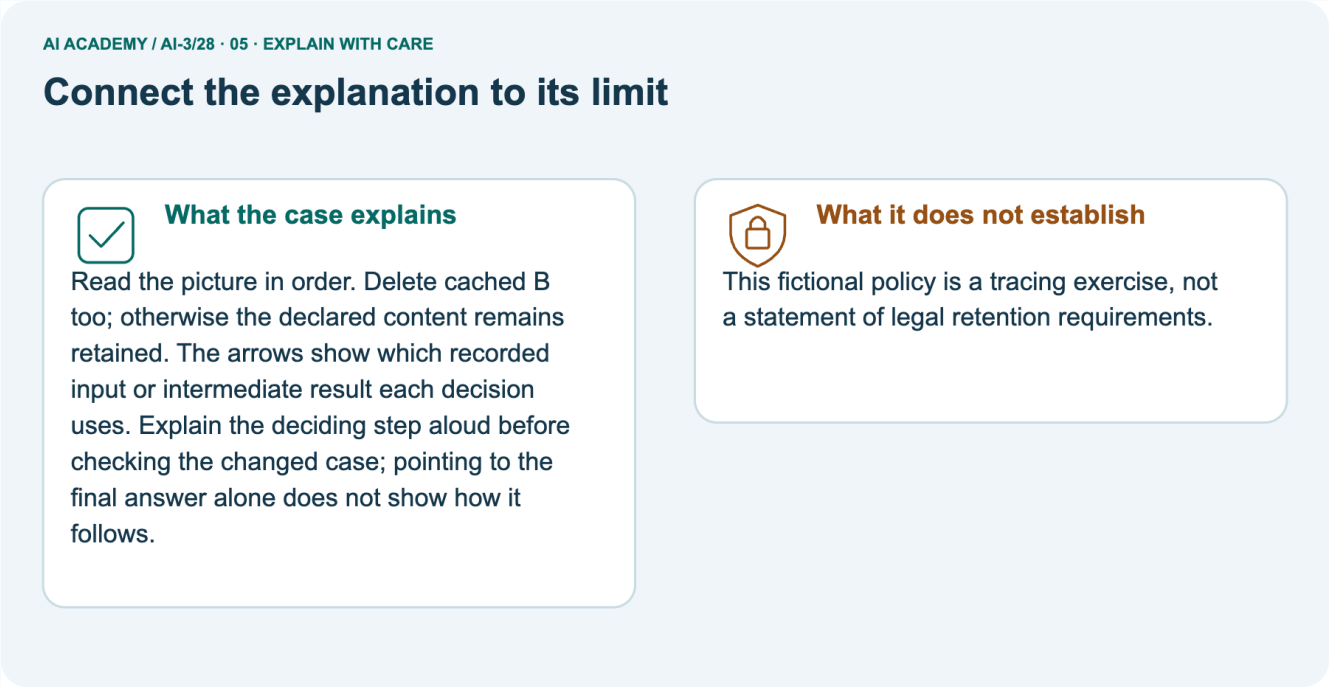
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Delete cached B too; otherwise the declared content remains retained. Changed condition: The cache deletion fails. Expected result: Report incomplete deletion rather than success.

Reason: Removing the working copy does not remove the known retained cache copy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

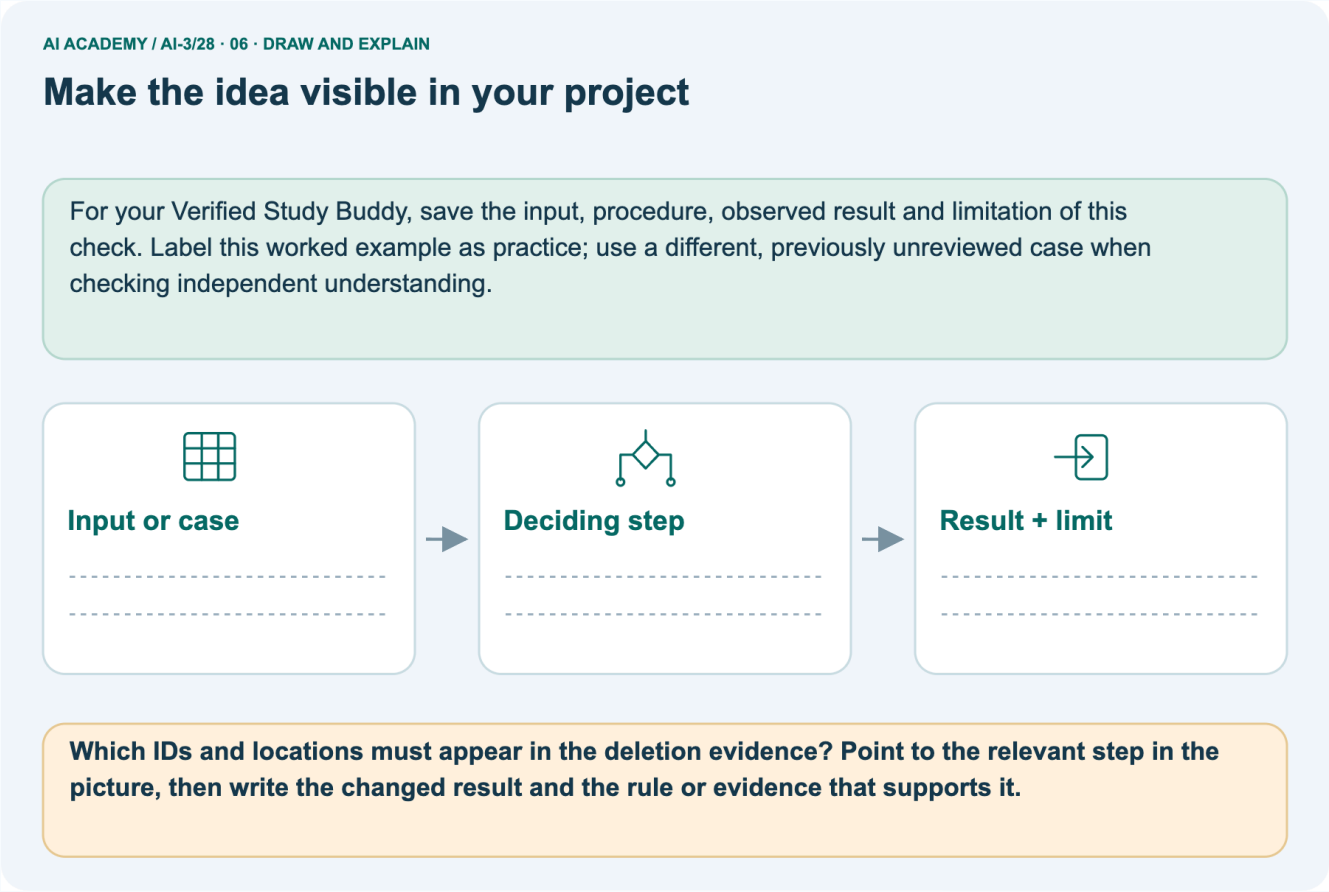
Read the picture in order. Delete cached B too; otherwise the declared content remains retained. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This fictional policy is a tracing exercise, not a statement of legal **retention** requirements.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which IDs and locations must appear in the deletion **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Privacy by Design: the case

02. Follow the deciding steps

03. Data retention and deletion — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.